From Uncertainty to the Schrödinger Equation

How Quantum Mechanics Starts Describing the Electron

Quantum Chemistry Notes
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Big Idea

The early quantum problems showed that classical physics was incomplete. But now we need a proper mathematical language to describe electrons. That language begins with matter waves, uncertainty, wavefunctions, and finally the Schrödinger equation.

1 Where are we now?

In the previous note, I looked at why quantum mechanics was needed in the first place.

Classical physics failed to explain blackbody radiation, the photoelectric effect, atomic stability, and line spectra.

But after seeing all those failures, the next question is natural:

Question

If electrons are not tiny planets orbiting the nucleus, then what are they?

This is where quantum mechanics starts becoming a proper theory.

The early quantum ideas told us that energy comes in packets and that atoms have discrete energy levels. But that is still not a complete description of the electron.

To describe the electron properly, we need a new idea of what a physical state even means.

In classical mechanics, the state of a particle can be described using position and momentum.

For example, if I know where a ball is and how fast it is moving, I can predict what happens next.

But for an electron, this picture breaks.

Quantum mechanics replaces the classical state with a new object:

the wavefunction

Before reaching the wavefunction and the Schrödinger equation, we first need to understand why particles started being treated like waves.

2 A short historical bridge

Around 1925, quantum mechanics began taking its modern form.

Heisenberg developed **matrix mechanics**, which described quantum systems using observable quantities like transition frequencies and intensities.

Soon after, Schrödinger developed **wave mechanics**, where quantum systems were described using wavefunctions.

At first, these approaches looked very different. Later, it was shown that they were mathematically equivalent.

Note to Self

For these notes, I am mainly following the wave mechanics route because it connects more naturally to chemistry, orbitals, and the Schrödinger equation.

3 Matter waves

3.1 The reverse question

The photoelectric effect showed that light, which was thought to be a wave, can behave like particles.

So de Broglie asked the reverse question:

Question

If waves can behave like particles, can particles behave like waves?

His idea was that every particle has a wavelength.

This wavelength is called the **de Broglie wavelength**.

$$\lambda = \frac{h}{p}$$

where:

- λ is the wavelength,
- h is Planck's constant,
- p is momentum.

Since wave number is defined as:

$$k = \frac{2\pi}{\lambda}$$

we can also write momentum as:

$$p = \hbar k$$

where:

$$\hbar = \frac{h}{2\pi}$$

Big Idea

The electron is not just a tiny particle. It has wave-like behavior, and its momentum is connected to its wavelength.

3.2 Why this matters

For large objects, the de Broglie wavelength is extremely tiny. That is why we do not notice wave behavior in footballs, cars, or humans.

But for electrons, the wavelength is not negligible.

This is why electrons can show diffraction and interference, just like waves.

That already tells me something important:

The electron cannot be fully understood as a classical particle.

4 The uncertainty principle

4.1 The statement

The uncertainty principle says that there is a fundamental limit to how precisely certain pairs of physical properties can be known at the same time.

The most famous pair is position and momentum.

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

where:

- Δx is the uncertainty in position,
- Δp is the uncertainty in momentum,
- $\hbar$ is the reduced Planck's constant.

This means:

The more precisely I know position, the less precisely I can know momentum.

And the reverse is also true:

The more precisely I know momentum, the less precisely I can know position.

4.2 What it does not mean

A common misunderstanding is that uncertainty is just because our instruments are bad.

That is not the real point.

It is not only a measurement problem.

It is built into the wave nature of quantum objects.

Note to Self

The uncertainty principle is not saying, "we are not smart enough to measure better." It is saying that quantum objects do not naturally have sharply defined position and momentum at the same time.

5 Connecting uncertainty to waves

This is the part that made the most intuitive sense to me.

If a particle has wave nature, then its momentum is connected to its wavelength.

$$p = \frac{h}{\lambda}$$

So if I know the wavelength very precisely, I know the momentum very precisely.

But a wave with one exact wavelength is spread out everywhere.

It does not have a well-defined position.

So:

Definite wavelength means definite momentum, but uncertain position.

Now imagine I want to localize the electron.

To make the electron appear in a small region, I cannot use just one perfect wave. I need to add many waves of different wavelengths together.

This creates a wave packet.

The wave packet is more localized in space, so I know the position better.

But because I used many wavelengths, the momentum becomes less definite.

So:

Better position information means worse momentum information.

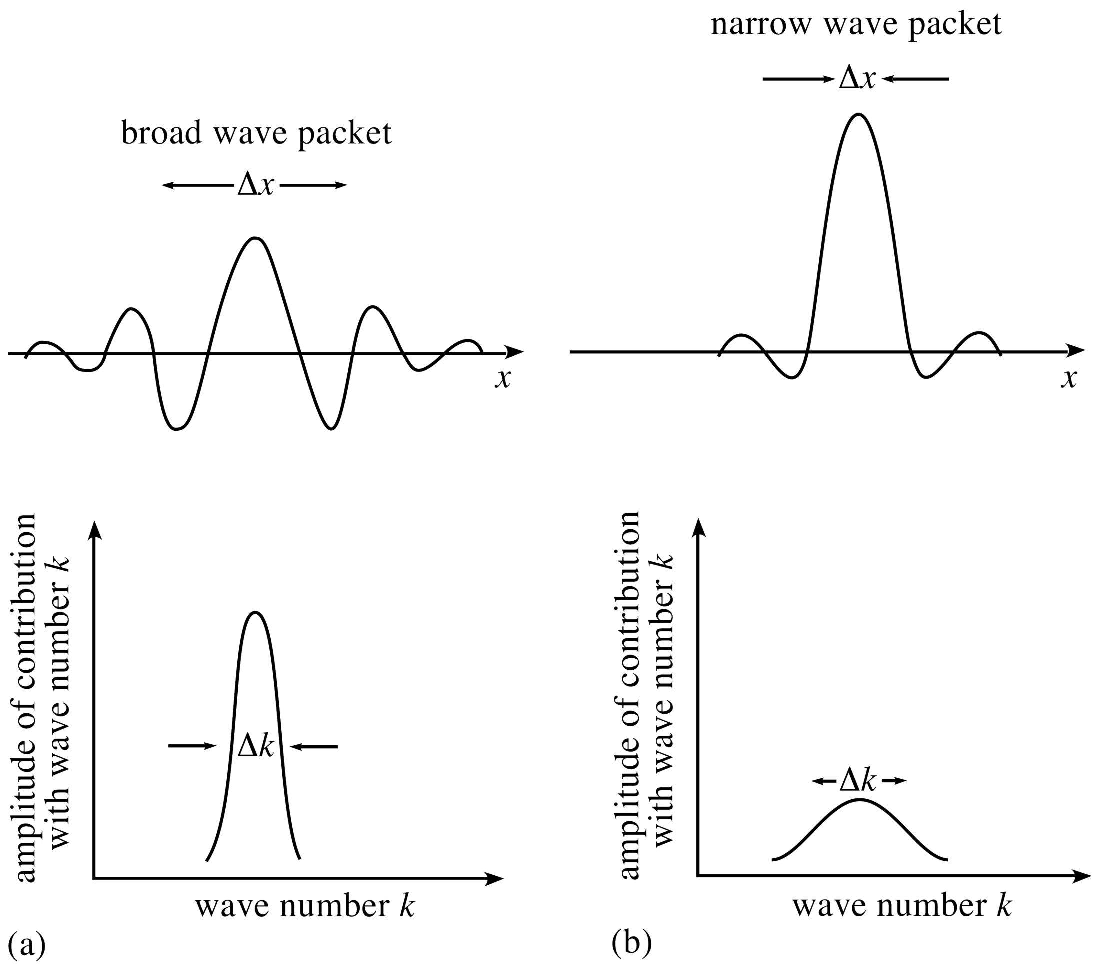

Figure 1: A broader wave packet in position space corresponds to a narrower packet in momentum space, while a more localized position packet corresponds to a broader momentum packet. Image source: Physics Stack Exchange, *Is this explanation for the uncertainty principle correct?*, https://physics.stackexchange.com/questions/479931/is-this-explanation-for-the-uncertainty-principle-correct.

Big Idea

Uncertainty is not random magic. It comes naturally from describing particles as waves. A localized wave needs many wavelengths, and many wavelengths mean many possible momenta.

6 The wavefunction

6.1 The basic definition

The wavefunction is usually denoted by:

$$\psi(x)$$

or, if time is included:

$$\psi(x,t)$$

A basic definition is:

The wavefunction is the mathematical object that describes the quantum state of a system.

But that sentence feels too dry.

So I think of it like this:

Big Idea

In classical mechanics, the state of a particle is described by position and momentum. In quantum mechanics, the state is described by the wavefunction.

The wavefunction contains the information we can use to calculate possible measurement results.

For example, from the wavefunction we can find:

- probability of finding the particle in a region,
- energy information when we apply the Hamiltonian operator,
- expectation values of observables,
- how the system evolves with time.

6.2 Is the wavefunction physical?

This is a subtle question.

The wavefunction is not physical in the same direct way that a ball or a table is physical.

We do not simply see the wavefunction.

But it is not meaningless either.

The most useful interpretation for now is the Born interpretation:

$$|\psi(x,t)|^2$$

This gives the **probability density** of finding the particle near position x at time t.

So in one dimension, the probability of finding the particle between a and b is:

$$P(a \leq x \leq b) = \int_a^b |\psi(x,t)|^2 \, dx$$

Note to Self

The wavefunction itself can be complex, but probability must be real and positive. That is why we use $|\psi|^2$.

7 Normalization

If $|\psi(x,t)|^2$ gives probability density, then the total probability of finding the particle somewhere must be 1.

For a particle in one dimension:

$$\int_{-\infty}^{\infty} |\psi(x,t)|^2 \, dx = 1$$

This is called **normalization**.

If a wavefunction is not normalized, we can often normalize it by multiplying it by a constant.

$$\psi_{\text{norm}}(x) = A\psi(x)$$

where A is chosen so that:

$$\int_{-\infty}^{\infty} |\psi_{\text{norm}}(x)|^2 \, dx = 1$$

7.1 Same physical state

If two wavefunctions differ only by a non-zero constant factor, they often represent the same physical state after normalization.

For example:

$$\phi(x) = \alpha\psi(x)$$

where α is a non-zero complex number.

After normalization, they give the same probability distribution.

Also, multiplying a normalized wavefunction by a pure phase does not change the probability density:

$$\psi(x) \to e^{i\theta}\psi(x)$$

because:

$$|e^{i\theta}\psi(x)|^2 = |\psi(x)|^2$$

Note to Self

The absolute size or global phase of a wavefunction is not usually the physical thing. What matters are probabilities, relative phases, and how states combine.

8 Superposition

One of the most important ideas in quantum mechanics is superposition.

If $\psi_1(x)$ and $\psi_2(x)$ are possible wavefunctions, then a combination like:

$$\psi(x) = c_1\psi_1(x) + c_2\psi_2(x)$$

can also be a possible wavefunction.

This means quantum states can be added together.

But this does not mean we can casually change one part of the wavefunction without changing the physics.

For example:

$$\psi(x) = \phi(x) + \omega(x)$$

is generally not the same physical state as:

$$\tilde{\psi}(x) = \alpha\phi(x) + \omega(x)$$

unless the whole wavefunction is multiplied by the same constant.

The relative size and relative phase between parts of a superposition matter.

Big Idea

A global phase usually does not change the physical state. But relative phase between parts of a superposition can change the physics.

9 Operators and observables

In quantum mechanics, measurable quantities are called **observables**.

Examples include:

- position,
- momentum,
- energy,
- angular momentum.

In classical mechanics, position and momentum are just numbers.

In quantum mechanics, observables are represented by **operators**.

For example, in one dimension:

$$\hat{x} = x$$

and momentum is represented by:

$$\hat{p} = -i\hbar \frac{d}{dx}$$

The energy operator is especially important.

It is called the **Hamiltonian**:

$$\hat{H}$$

For a particle moving in one dimension under a potential $V(x)$, the Hamiltonian is:

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$$

The first term represents kinetic energy.

The second term represents potential energy.

Connection

This is where atomic physics starts becoming visible. If we know the potential energy acting on an electron, we can build the Hamiltonian and solve for the allowed wavefunctions and energies.

10 The Schrödinger equation

10.1 Why we need it

If electrons behave like waves, then we need an equation that tells us how the wavefunction behaves.

That equation is the Schrödinger equation.

The time-dependent Schrödinger equation is:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

This tells us how the wavefunction changes with time.

Big Idea

The Schrödinger equation is not mainly telling us where the electron is. It tells us how the wavefunction evolves.

10.2 The time-independent version

In many quantum chemistry problems, we focus on stationary states.

For those cases, we use the time-independent Schrödinger equation:

$$\hat{H}\psi = E\psi$$

This equation says:

Find the wavefunctions that have definite energy.

Here:

- $\hat{H}$ is the Hamiltonian operator,
- ψ is the wavefunction,
- E is the energy of that state.

This is an eigenvalue equation.

The allowed wavefunctions are called eigenfunctions.

The allowed energies are called eigenvalues.

10.3 What it really means

The equation:

$$\hat{H}\psi = E\psi$$

looks small, but it is huge in meaning.

It says that only certain wavefunctions fit the system.

And each allowed wavefunction has a corresponding energy.

This is where quantization appears naturally.

For bound systems, energy levels are not randomly assumed anymore. They come from solving the Schrödinger equation with the correct boundary conditions.

Note to Self

This is powerful because the earlier quantum theory had to guess allowed energy levels. Schrödinger's equation gives a systematic way to find them.

11 A simple mental picture

I should not imagine the electron as a tiny ball secretly moving along a hidden path.

For now, it is better to think of the electron's state as a wave-like object.

The wavefunction can spread, interfere, and form standing-wave-like patterns.

When the wavefunction is trapped by a potential, only certain stable patterns are allowed.

This is similar to how a guitar string can vibrate only in certain modes.

In atoms, the allowed electron wavefunctions become orbitals.

Big Idea

Orbitals are not paths followed by electrons. They are allowed wavefunction patterns for electrons in atoms and molecules.

12 How this connects to atomic stability

In the previous note, atomic stability was one of the biggest failures of classical physics.

Classically, the electron should spiral into the nucleus.

But quantum mechanically, the electron is described by a wavefunction.

When we solve the Schrödinger equation for the hydrogen atom, we find that the electron has allowed energy levels.

The lowest one is the ground state.

There is no lower energy state for the electron to keep falling into.

Also, trying to confine the electron too close to the nucleus increases its momentum uncertainty, which increases its kinetic energy.

So stability comes from the quantum structure of the wavefunction.

Connection

The uncertainty principle and Schrödinger equation are not separate random ideas. Together, they explain why the electron does not collapse into the nucleus.

13 Quick Summary

Idea	Meaning	Why it matters
Matter waves	Particles have wavelength, $\lambda = h/p$	Electrons cannot be treated only as classical particles
Uncertainty principle	$\Delta x \Delta p \geq \hbar/2$	Position and momentum cannot both be perfectly sharp
Wavefunction	The quantum state of a system	Gives probability information and physical predictions
Normalization	Total probability must equal 1	Makes $\lvert\psi\rvert^2$ usable as probability density
Superposition	Quantum states can be added	Explains interference and mixed states
Schrödinger equation	Equation for wavefunction evolution and energy states	Foundation for orbitals, spectra, and bonding

Table 1: The main ideas that lead from uncertainty to the Schrödinger equation.

14 Final Takeaway

The failures of classical physics forced physicists to rethink the microscopic world.

But quantum mechanics needed more than just the idea of energy packets.

It needed a new way to describe particles.

Matter waves showed that electrons have wave-like behavior.

The uncertainty principle showed that position and momentum cannot both be perfectly defined.

The wavefunction became the object that describes the quantum state.

The Schrödinger equation became the equation that tells us how the wavefunction behaves.

For studying atoms, orbitals, and eventually molecules, this is the real entry point.

References and Reading Notes

- HyperPhysics, notes on the uncertainty principle.
- Carnegie Mellon Quantum Theory notes, discussion of wavefunctions and physical states.
- Physics Stack Exchange, *Is this explanation for the uncertainty principle correct?*, `https://physics.stackexchange.com/questions/479931/is-this-explanation-for-the-uncertainty-principle-correct`.
- Standard quantum mechanics texts on de Broglie matter waves, wavefunctions, and the Schrödinger equation.